



Influence of solar radiation on the surface loss of the painted statues in Grotto No. 3 of Maijishan Grottoes, China

Ruobin Wu^a, Yan Ma^a, Huarong Xie^a, Xudong Wang^b, Shuichi Hokoi^a, Yongqiang Yue^c, Junjian Hu^c, Baogang Mu^d, Yonghui Li^{a, e, *}

^a School of Architecture, Southeast University, 210096, Nanjing, PR China

^b The Palace Museum, 100009, Beijing, PR China

^c Art Institute of Maijishan Cave-Temple Complex, 741020, Tianshui, PR China

^d School of Civil Engineering, Southeast University, 211189, Nanjing, PR China

^e Key Laboratory of Urban and Architectural Heritage Conservation of Ministry of Education (Southeast University), 210096, Nanjing, PR China

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ABSTRACT

The embedded hygrothermal environment of cultural heritages (solar radiation, temperature, humidity, etc.) plays a significant role in their conservation and survival, especially in the case of outdoor ones. The surface loss of painted statues is often related to fluctuations in the surface temperature caused by the ever-changing solar radiation; however, the quantitative relationship between them remains unclear. To clarify, in this paper, hygrothermal monitoring and numerical simulation were conducted on painted statues with an age of over 1400 years in the World Heritage Maijishan Grottoes. The results showed that the ruined eaves have led to significant differences in solar radiation intensity and surface temperature between the upper and lower statues, with maximum values of 598.7 W/m² and 13.6 °C, respectively, corresponding to surface loss rates of nearly 15% and 50%. Under the influence of direct insolation, the thermal stress on the statue surface increased with decreasing height, resulting in a difference of 10.8 kPa between the highest and lowest points. Taking the tensile strength of clay (10 kPa) as a threshold of thermal stress, the yearly number of events exceeding this threshold at the upper and lower rows were 2819 and 3173, respectively. Overall, direct insolation has a significant effect on the temperature changes on painted statue surfaces, thus leading to greater thermal stress and greater degrees of surface loss. For heritages in open or semi-open environments, it is necessary and effective to evaluate their preservation conditions based on the threshold and then propose suitable conservation methods for preventive protection.

1. Introduction

Temperature and moisture content changes, salt efflorescence and subflorescence, freeze-thaw damage, etc., in the natural environment are thought to be the main drivers of physical weathering on cultural heritages [1–4], as well as being pivotal conditions for other types of degradation [5–7]. When weathering takes place on cultural heritage, their historical, cultural, and aesthetic value will be eroded, causing irreversible damage, and even threatening the survival of the heritage [8,9].

Temperature is one of the key factors informing the degradation of materials, and direct insolation (solar radiation) is the main reason for temperature changes in the natural environment [10]. For example, repeated and rapid changes in direct insolation over the course of a day cause different thermal responses within the material [11]. Studies

have reported that under such circumstances, changes in surface temperature will superimpose different time scales (from minutes to hours) upon diurnal patterns, sometimes even at speeds greater than 3 °C/min [12]. Therefore, areas near the surface, at a depth of about 10 mm, on rock show greater weathering than the inside [13], and it has also been confirmed by B. J. Smith et al. that huge temperature and stress differences exist between the surface and inner areas of exposed stone [14]. The amount of direct insolation is dependent not only on the duration, but also on the solar altitude, azimuth, and other conditions (such as cloud occlusion, wind, etc.) [15], and thus, only a limited depth is affected by the daily cycle. Many studies have examined thermal responses or mechanisms of materials [16–19]. The heterogeneity inside one material or structure, such as lineal expansion ratio, textural characteristics, etc., may contribute to uneven internal thermal deformation, leading to stress concentration and failure [20–24]. For example,

* Corresponding author. #2 Sipailou, Xuanwu District, 210096, Nanjing City, Jiangsu Province, PR China.

E-mail address: liyonghui@seu.edu.cn (Y. Li).

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the monitored and simulated results in St Vitus' Cathedral exhibited dynamic deformation of the bearing structure in different seasons [25], as did the Angkor Wat Temple [26]. As to the micro-level, the differences in micro-texture in a forty-two-thermal-cycle experiment revealed that substantial changes occurred in the granite, which may continue to develop over time [27].

The temperature fluctuation under direct insolation intensifies changes in migration of water and salt, which is one of the main driving forces in the deterioration of heritages [28,29]. Solar-induced temperature changes (thermal stresses) in the natural environment may participate in virtually all weathering processes, thus causing the development of sub-critical cracks [30]. Specifically, they can activate or accelerate weathering in synergy with other factors, such as the processes of rainfall, freeze–thaw, salt crystallization [31], etc., thus forming a vicious cycle [32–34]. For example, the percentile of thermal stress distribution on both the south and west sides of the Arch of Augustus are the same, but more factors are intertwined on the south side, resulting in a more rapid deterioration rate [35]. A similar phenomenon is found at the Sassi di Matera site, as well [36]. In Angkor Wat, more weathering of the outer colonnades than the inner also shows the spatial distribution characteristics of environmental factors (precipitation and solar radiation) [37]. This is because the superimposed thermal response on diurnal patterns aggravates the transport of moisture and salt on the surface area [19], causing damage to the internal pore structure [38].

Statues in grotto temples or in similar preservation environments are common and can be found worldwide, with most of them having precious legacies. As one of the most valuable types of artifact found in grotto temples, statues are often protected with wooden or stone cave eaves or transition spaces created when the grottoes were carved [39]. However, most of these ancient protective measures have been damaged or have even disappeared over time, leaving the statues directly exposed to solar radiation, wind-driven rain, etc., thus causing further deterioration [40,41] and the loss of their value. Like murals, stone-based statue consists of a combination of roughly carved rock as a support, clay as the earthen plaster layer and colorful drawings (surface pigments) as paint layers. The different properties of the multilayer structure (such as thermal and hygroscopic expansion, etc.) have been found to be a critical factor leading to the failure between the layers [42,43], as well as between the surface pigments and clay [44]. Previous studies have discussed the deterioration and conservation of cultural heritages from the perspective of coupled heat and moisture transfer; however, few have discussed the role of solar radiation and thermal stress. In fact, the change in thermal stress is an important reason for the flaking or loss of painted statues with important historical value [45,46].

In this study, the influence of thermal stress is introduced to evaluate the surface loss of painted statues. Given the difference in the deterioration of two adjacent rows of Buddha statues with a long history of over 1400 years, the semi-open grotto (Grotto No. 3) of the Maijishan Grottoes was selected to conduct this research, in which direct insolation was recognized as an important factor. Thus, a monitoring scheme against solar radiation, an investigation of deterioration through surface loss, and two-dimensional heat and moisture transfer modeling were executed to analyze the correlation between solar radiation and the loss rate and to evaluate its contribution to the difference in deterioration. To prevent the occurrence of damage from thermal stress, an attempt was made to figure out a threshold, as well as possible protection schemes. The results may deepen the understanding of the effects of solar radiation and thermal stress, and they may have guiding significance for preventive protection methods and the design of heritages exposed to the environment.

2. Materials and methods

2.1. Heritage profile

The Maijishan Grottoes (N: 34°35'; E: 106°00') were announced as a World Cultural Heritage Site in 2014 by UNESCO in Gansu Province, China. The earliest grottoes can be traced back over 1600 years (to the Northern Wei Dynasty). There are 221 grottoes, over 7800 statues, and nearly 1000 square meters of murals remaining [47]. The grottoes are attached to the Danxia landform (Maiji Mountain), whose lithology is Tertiary brick-red thick massif conglomerate, sand conglomerate, and sandy mudstone. Carved on the south façade of Maiji Mountain, lots of the grottoes and artifacts inside are open or semi-open to the outdoor environment. As many of the statues are carved on the rock (stone-based statues) and are multilayered structure (clay as earthen plaster layer and colorful drawings as paint layers), their direct exposure to strong solar radiation and violent fluctuations in surface temperature may play an important role in the surface deterioration, which can also be found in the case of other artifacts at this location.

Grotto No. 3 is one of the semi-open grottoes carved in the middle of the southeast façade of Maiji Mountain (Fig. 1a). It is 36.5 m long and is at a height about 50–60 m from the ground. There are six rows of Buddha statues (nearly three hundred in total), which are all stone-based statues, whose thickness of the earthen plaster layer is about 2 cm [48] (Fig. 1b). The upper two rows could date back to the Northern Zhou Dynasty (over 1400 years), and were repaired and redecorated in the Qing and Song Dynasties, while the lower four rows were sculpted later, during the Sui Dynasty. According to a field survey, the 3-m outer front eaves have collapsed and have been damaged unevenly, leaving the statues inside exposed to direct insolation in an uneven manner. Many kinds of deterioration of the statues were observed, including flaking, craquelure (Fig. 1c), color fading, hollow (detachment of earthen plaster later; Fig. 1d), deep loss (Fig. 1e), overlapping of pigment layers (Fig. 1f), cracks on the rock (Fig. 1g), etc., some of which may be strongly related to the sunlight. Moreover, obvious differences in deterioration between the statues at the upper two rows were observed. Therefore, understanding the influence of solar radiation on the surface loss of Buddha statues will play a positive role in supporting and realizing the protection of grotto temples.

2.2. Monitoring scheme

To clarify the relationship between solar radiation and deterioration, a monitoring scheme against solar radiation difference was designed (Fig. 2a–d). The monitored area was the upper two rows of statues in Grotto No. 3 (above the walkway), where significant differences in the loss area between the two rows of statues was observed. Both rows are exposed to direct insolation, while the sunlight is distributed unevenly due to the ruined eaves. Therefore, two vertical radiation sensors were set in the same direction and at the same height as the heads of the two statues (Fig. 2e). Three K-type thermocouples were placed at different heights (denoted as the upper, middle, and lower points) to measure the surface temperature. In consideration of minimizing the impact on the statues or the drawings (risk of damage and changes in appearance) and keep the similar hygrothermal and radiation behavior to surrounding Buddha surface, the temperature sensors were applied to the bedrock with desalted clay that was also used to repair the murals. To secure the robust of the thermocouples, the thickness of the clay is about 1–2 cm after installation (Fig. 2f). For ambient air temperature

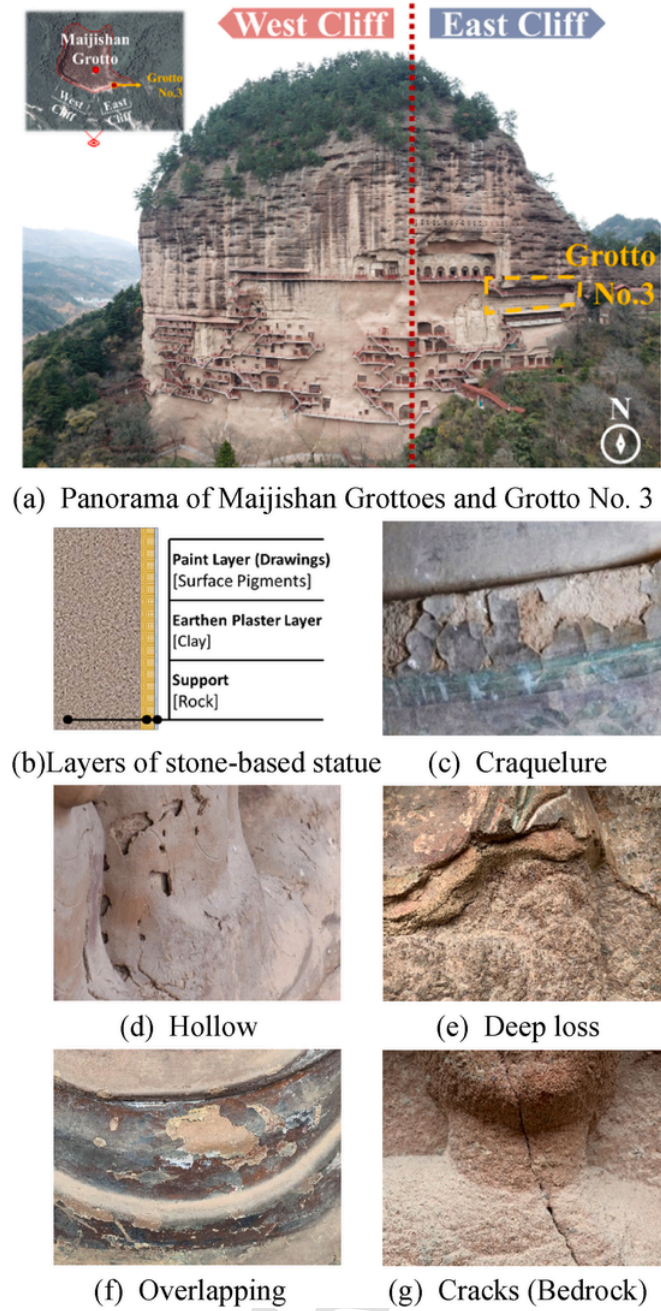


Fig. 1. Maijishan Grottoes, Grotto No. 3 and current damage types in the grotto.

and relative humidity, the monitoring data are retrieved from the weather station of the monitoring and early warning platform in the Maijishan Grottoes. Detailed information on the pyranometers, temperature and relative humidity sensors and weather station is listed in Table 1. All were recorded at an interval of 10 min. The monitoring data will serve as the basis and input data for the validation and calculation of the hygrothermal model.

2.3. Hygrothermal model

To evaluate the influence of solar radiation on the thermal stress of the statue surface, a prediction model was established using the coupled heat and moisture transfer theory. It was proposed by Matsumoto and has been continuously developed in the past decades [49,50]. This model has been widely applied to predict and assess the deterioration of cultural heritages, such as Hagia Sophia [45], Takamatsu Tumulus

[51,52], stone carvings [53], etc. [28,52,54]. The model characterizes temperature and water chemical potential as driving forces and was compiled using the Fortran language on the Intel Visual Studio platform.

The chemical potential of water is related to temperature and relative humidity, as shown in Equation (1). The heat and moisture balance equations are shown as Equations (2) and (3), respectively (the notation is explained in the Appendix).

$$\mu = R_v T \ln(RH) \quad (1)$$

$$c\rho \cdot \frac{\partial T}{\partial t} = \nabla \cdot \left[\left(\lambda + \lambda'_{Tg} \right) \cdot \nabla T + r \cdot \lambda'_{\mu g} \cdot \nabla \mu \right] \quad (2)$$

$$\rho_{water} \cdot \frac{\partial \Psi}{\partial \mu} \cdot \frac{\partial \mu}{\partial t} = \nabla \cdot \left[\lambda'_T \cdot \nabla T + \lambda'_\mu \cdot (\nabla \mu - \bar{n}g) \right] \quad (3)$$

In the simulation model, a north-south section (15° south by east) and simplified outlines were used for calculation. Thus, a rectangular coordinate system is established, with the horizontal X-axis to the right and the vertical Y-axis upward. The diagram of the model is shown in Fig. 3a and b. Four kinds of material or areas were distinguished: the rock as support (grey part), the Buddha clay as earthen plaster layer (red part), the concrete walkway (blue part), and the outdoor air (blank part). The computational grids were intentionally divided into different densities in consideration of calculation accuracy and efficiency. The finite difference method was used with a calculation time step of 30 s.

The data of ambient air temperature, relative humidity, and horizontal solar radiation from September 1, 2020 to August 31, 2021 recorded by the weather station near Maiji Mountain, were input as the boundary conditions. The boundaries of the calculation model are shown in Fig. 3a. The detailed distribution of materials and computational grids around Grotto No. 3 is shown in Fig. 3b. The left boundary (15 m deep from the surface of the mountain) is considered constant at 10.2 °C, the annual average outdoor air temperature. The top boundary is 15 m away from the edge of the grotto eaves, and similarly, the bottom is the same distance away from the concrete walkway. Neumann boundary conditions are set at both the top and bottom boundaries. To identify spatial differences in the solar radiation on and the surface temperature of the statues (206 cm from the bottom of the Buddha Clay in the lower row to the top of the upper row, Fig. 3c), it is divided into parts with an interval of 10 cm between H10 and H90 and between H110 and H200 (with the interval between H90 and H110 of 20 cm). The points at heights of 10, 20, ..., 190, 200 cm were designated as H10, H20, ..., H190, H200. Bouguer's equation (Equation (4)) and Berlage's equation (Equation (5); modified by Matsuo Yoh [55]) are used as the calculation model of solar radiation [45,56,57], with the equivalent effects of weather conditions (clouds, atmospheric transmission, etc.) being considered by comparing the monitored and theoretical vertical solar radiation of the two rows. On this basis, we calculated the direct and diffuse solar radiation, as well as that in other directions. The hours of direct sunlight were calculated on the basis of the solar altitude and azimuth, and the yearly total amount of radiation was obtained by adding the received solar radiation per square meter per hour.

$$J = J_0 \cdot P^{cosec \theta} \quad (4)$$

$$J_{sh} = 1.2 \cdot J_0 \cdot \sin \theta \cdot \frac{(1 - P) \cdot (1 - P^{cosec \theta})}{(1 - 1.4 \cdot \ln P)} \quad (5)$$

The convection and radiation heat transfer coefficients were calculated using empirical formulas (Equation (6)–(8)) [58]:

$$h_c = a \cdot \sqrt[4]{T_{surface} - T_{out}} + 4.2v_{wind} \quad (6)$$

$$h_r = \frac{4 \cdot C_{1,2} \cdot T_{surface}^3}{10^8} \cdot \bar{\psi}_{12} \quad (7)$$

$$h_{total} = h_c + h_r \quad (8)$$

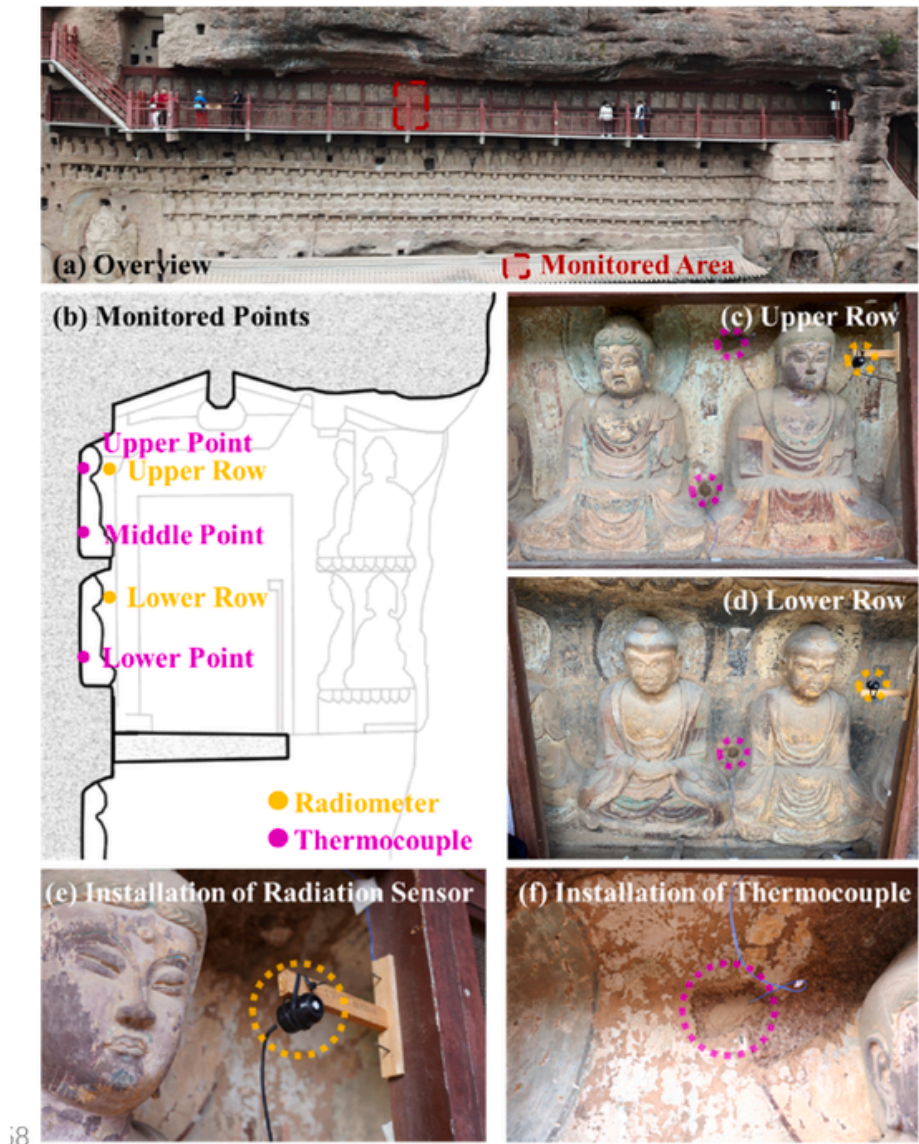


Fig. 2. Monitoring scheme of Grotto No. 3.

Table 1
Details of monitoring equipment.

	Temperature Recorder	Radiation Sensor	Temperature	Relative Humidity	Precipitation
Type	TESTO 176 T 4 (Thermocouple)	S-LIB-M 003 (with H 21-USB)	LUFFT WS600-UMB Smart Weather Sensor		
Range	−195–1000 °C (K-type)	0–1280 W/m ²	−50–60 °C	0–100%	0.5–200 mm/h
Accuracy	±0.3 °C (−100–70 °C)	±10 W/m ² or ±5%	±0.2 °C (−20–50 °C), otherwise ±0.5 °C	±2% RH	20% under laboratory conditions
Resolution	0.1 °C	1.25 W/m ²	—	—	0.01 mm/h

Due to the protruding rock eaves, no rain or wind-driven rain was observed in the field survey, and thus no precipitation was assumed on the surface inside the eaves.

The simulation was executed and run circularly by using the input data. The difference of temperature and moisture content between the results of 10th cycle and the next cycle is within 0.22%. Therefore, the

calculated results of the 10th cycle were considered stable and used for subsequent analysis and evaluation.

2.4. Calculation of thermal stress

Thermal stress is generated when the deformation (in length or volume) caused by temperature changes interacts with the internal or external constraints of materials. In this paper, it is assumed that the materials are homogeneous and isotropic, and their linear thermal expansion coefficients (α) are constant. The linear expansion coefficient reflects the elongation or contraction per unit length per kelvin degree (K). To describe the thermal stress, a third Z-axis perpendicular to the XOY plane (Fig. 4) is introduced, forming a 3D Cartesian coordinate system. The statue is considered to be a semi-infinite medium with no stress along the X-axis ($\sigma_x = 0$) and no strain in the YOZ plane (statue surface; $\epsilon_y = \epsilon_z = 0$), potentially with thermal stress. Combining Hooke's law and the superimposed temperature-associated strains under temperature difference (ΔT), the thermal stress can be obtained using Equation (9) [59], in which T_1 is calculated on the basis of the hourly surface temperature (T_1) minus the annual average temperature (T_0). As the surface is affected the most (compared with the inside area) by the ambient hygrothermal environment, the surface temperature

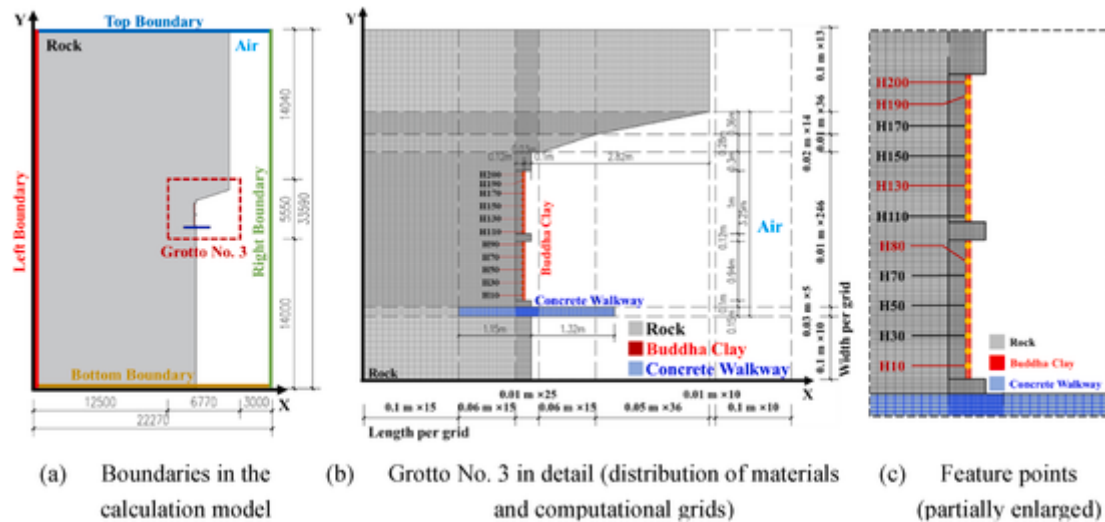


Fig. 3. The computational modeling of Grotto No. 3.

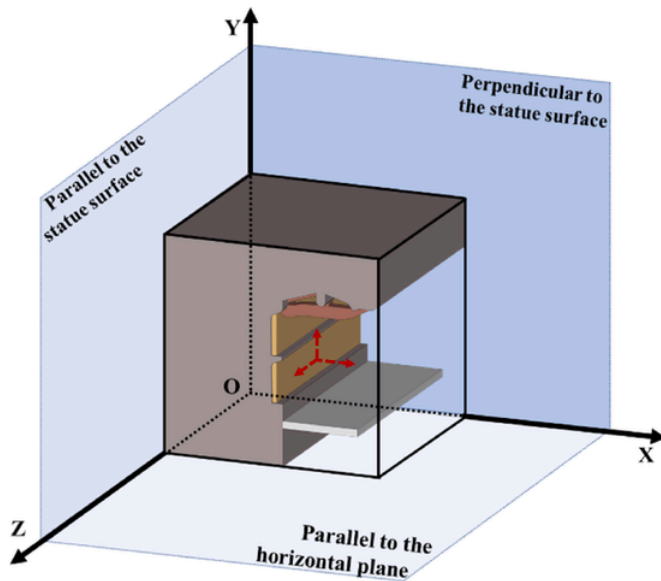


Fig. 4. The directions of thermal stress of the material.

(calculated by the hygrothermal model) was selected to represent the most unfavorable conditions. Regarding the annual average temperature, it is assumed that there is no stress at this temperature under annual periodic changes. The result of σ_T is specified as tensile stress when ΔT is positive and compressive stress when negative.

$$\sigma_T = \frac{E \cdot \alpha \cdot \Delta T}{1 - \nu} \quad (9)$$

Other mechanical parameters of the materials used for the calculation are listed in [Table 2](#). The elastic modulus (E) is a proportional coefficient of stress and strain during elastic deformation, showing the ability of one material to resist deformation (not strong resistance for the

Table 2
Parameters in the calculation of thermal stress.

Item	Clay
Elastic Modulus (E) [$\times 10^4$ MPa]	0.0887 [60]
Coefficient of Linear Expansion (α) [m/m·K]	1.2×10^{-6} [60]
Poisson's Ratio (ν) [—]	0.35 [61]

clay). As to Poisson's ratio, it characterizes transverse deformation of the material when being loaded within the elastic range.

2.5. Calculation of loss rate

In the field survey, the deterioration level of the upper and lower rows of statues in Grotto No. 3 differed. To quantify the differences, the loss of both earthen plaster (clay) and paint layers was counted using image analysis (pixel level). Therefore, the loss rates were obtained using the pixels of the loss area accounting for those of the whole statue. The loss of the earthen plaster layer refers to the damage to the integrity of the statue, for example, the lack of facial features, folds on the clothing, or other parts. In this case, the pigment (paint layer) on it is also lost. With respect to the paint layer, only areas falling off leaving the earthen plaster layer visible are counted. This is because the painting on the statues of the two rows is in multiple layers, having been redrawn throughout history. Thus, situations in which the overlying paint layer has peeled off while paint (e.g., the original paint layer) still remains on the earthen plaster layer are not included in the statistics.

3. Results

3.1. Loss rate of the statues

On the basis of the field survey and image analysis, the distributions of loss areas on the upper and lower rows of statues in Grotto No. 3 were obtained (Fig. 5). It can be observed that the total loss area (including both paint and earthen plaster layers) on the statue surface in the lower row (46.9%) is three times greater than that of the upper one (13.4%). Specifically, the loss rate of the earthen plaster layer in the lower row is 12.9%, while it is 1.3% in the upper one, which is almost ten times greater. The loss rates of the uppermost paint layer are 34.0% and 12.1% in the lower and upper rows, respectively. In addition, the earthen plaster loss rate of the lower statue alone is greater than the paint loss rate of the upper statue. Both rates show that the deterioration is more severe on the statues in the lower row.

3.2. Measured results

The one-year monitoring data (from September 1, 2020 to August 31, 2021) were obtained (Fig. 6). The measured results of outdoor air temperature and relative humidity, solar radiation intensity, and surface temperatures are as follows.

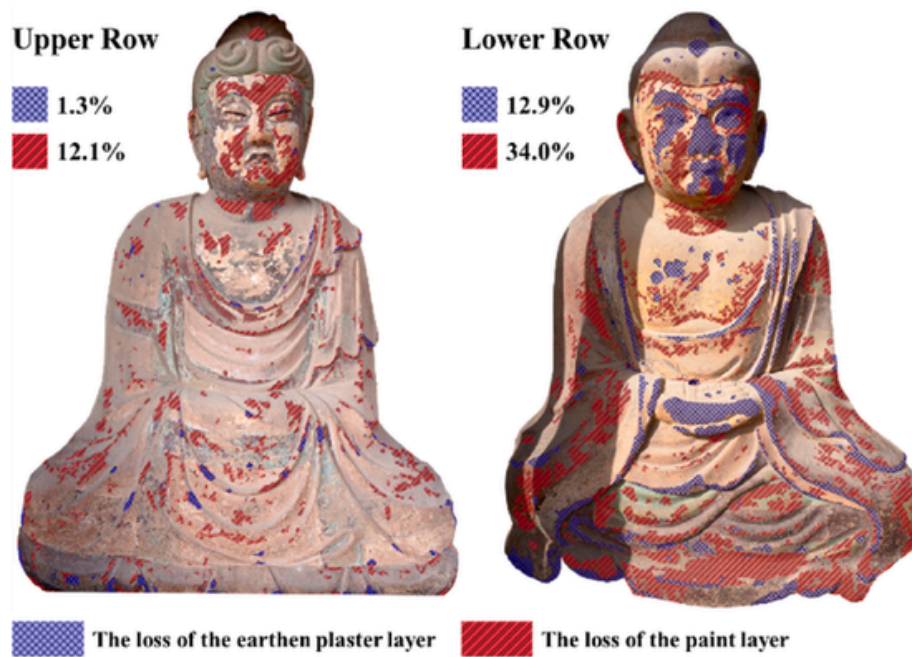


Fig. 5. The loss of paint and earthen plaster layers on the two rows of the statues in Grotto No. 3.

3.2.1. Air temperature and relative humidity

The variation in the air temperature in the Maijishan Grottoes during the monitored period is shown in Fig. 6a. The temperature fluctuates from -14.4°C to 38.7°C , with an annual average of 10.2°C . The coldest and hottest months are December (-3.1°C) and July (21.7°C), respectively. As to relative humidity, it varies from 11.6% to 100%, with the highest monthly average of 85.8% in October and the lowest of 54.3% in January and February. According to the statistics, there were 1455.2 h in which the temperature was below 0°C and 2008.8 h when the relative humidity exceeded 90%. Obvious seasonal differences in temperature and humidity were observed in this area, which is warm and humid in summer and cold and dry in winter.

3.2.2. Solar radiation intensity

Fig. 6b shows the solar radiation intensity monitored in the upper and lower rows in Grotto No. 3, both of which have similar but slightly different trends. The solar radiation intensity for both rows remained at a low level (around 100 W/m^2) from mid-March to mid-October, indicating exposure only to diffuse radiation, with no direct sunlight. In the other periods, the variation curves showed an increasing then a decreasing trend, with both peaks at 678 W/m^2 around the winter solstice (December 23rd). Overall, seasonal variations in solar radiation intensity can be observed in both rows, with direct insolation tending to impact the statue surfaces between October and March (influenced period).

The ruined eaves may contribute to the difference in the influenced time (duration) and intensity for the two rows. The seasonal variation in the lower row is almost one month earlier or later than that in the upper. For a given day, the time distribution of peaks during that day is also different, showing a delayed variation. The upper row receives sunlight earlier than the lower one; however, the latter reaches the same level or higher afterward. For example, sometimes the upper row was 300 W/m^2 higher than the lower one, while at other times the lower row was much higher (598.7 W/m^2 at most). Additionally, solar radiation intensity over 150 W/m^2 occurred for three times longer in the lower row (503.7 h) compared with the upper one (168.3 h). In conclusion, solar radiation intensity shows a spatial and temporal distribution in Grotto No. 3.

3.2.3. Surface temperature

Similar to the seasonal and diurnal variations in the monitored solar radiation intensity, the surface temperature in Grotto No. 3 at the upper, middle, and lower points (Fig. 6c) showed significant differences during the influenced period (October to March), while almost no difference was visible beyond this. Specifically, the daily variations in the surface temperatures at three points are different, varying from 0.5 to 7.2°C , from 0.4 to 14.0°C , and from 0.7 to 19.2°C , respectively. More intense daily fluctuations were found in the middle and lower parts of the statues, where higher solar radiation intensity tends to occur, and for longer. Under the influence of solar radiation, the surface temperature at the middle and lower points was nearly $8\text{--}14^{\circ}$ higher than that at the upper point. At the lower point, the maximum during winter even exceeded that in the summer, with a peak value of 26.5°C . In addition, for over 69% of the monitored time, the lower point has a temperature higher than or equal to the upper one, with a maximum temperature difference of 13.6°C . The number of hours for which the temperature change was greater than $2, 4, 6, 8, 10,$ and 12° was counted, revealing results of 655.7, 392.5, 294.0, 203.7, 104.2, and 19.2 h, respectively. As to the lower and middle points, the temperature at the lower point was equal to or greater than that at the middle one only 43.8% of the time, with a maximum temperature difference of 13.2°C . Similarly, the temperature change distribution from 2 to 12°C was 447.0, 209.2, 124.7, 79.5, 31.2, and 4.0 h, respectively.

In a word, the uneven and seasonal changes in solar radiation under the ruined eaves led to different temperature increases in terms of both daily variations and extremes on the statue surface, with the lower point being most affected, followed by the middle and the upper points.

3.3. Model validation

The model was calibrated sequentially for solar radiation intensity and surface temperature, in a form whereby the trend, values and correlation analysis could be compared between the monitored and the calculated results.

With respect to solar radiation intensity, similar trends were found at the two monitored points and the corresponding position in the numerical model (Fig. 7a, the lower row, for example). Then, the correlation analyses gave good fittings, with the R^2 values of 0.8228 at the

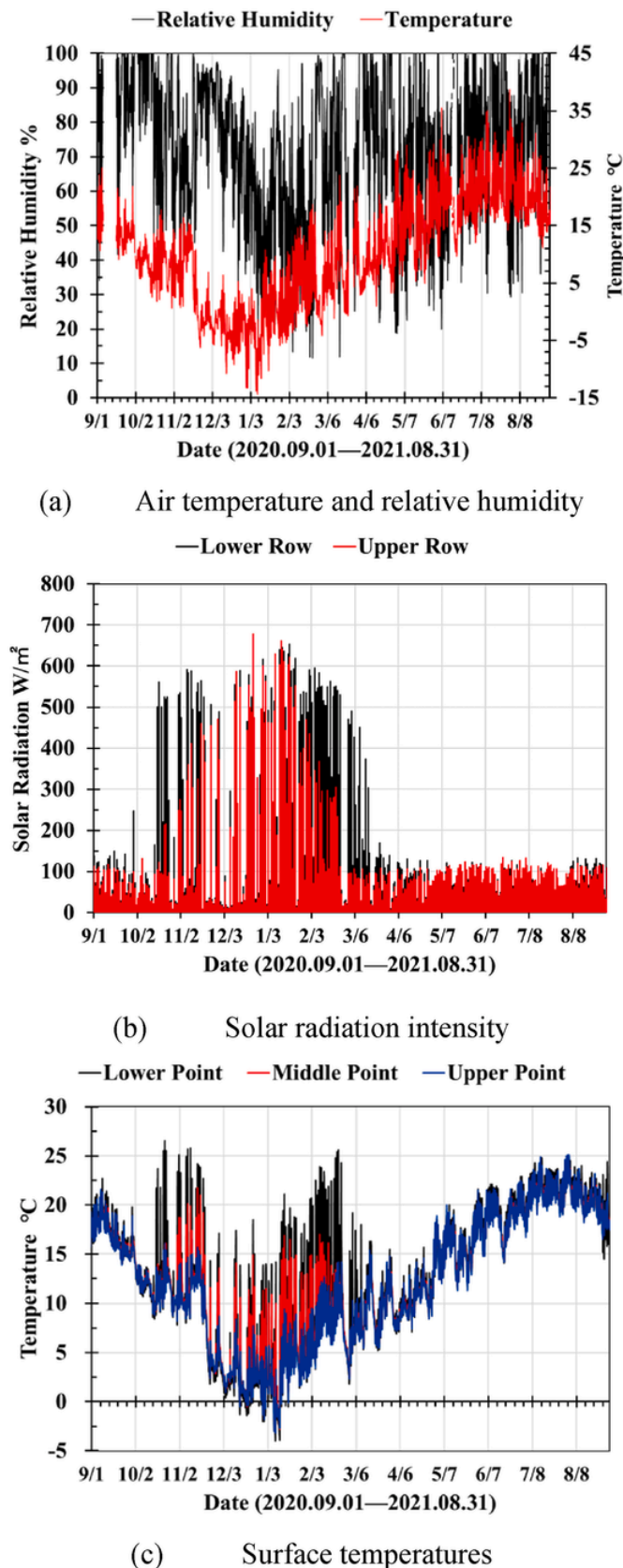


Fig. 6. Monitored results.

lower row and 0.9228 at the upper one, respectively. Moreover, detailed values showed that the differences were small enough to perform further temperature validation. For example, the difference between the calculated and monitored maximum values and the total amount on the lower row were 740.3 versus 678.1 W/m^2 and 291.97 versus 329.51 $kW h/m^2$, respectively. Then, the measured and calculated surface temperatures at three points were compared. Taking the lower point as an example (Fig. 7b), the variations in both values were similar, but in the simulation, the minimum temperature was slightly lower, and the daily amplitude was larger. The R^2 values of the three points from top to bottom were 0.9620, 0.9383, and 0.9609, respectively.

Although differences between measured and calculated values can be seen, for example, possible overestimation brought by few times of higher calculated solar radiation intensity in winter, the overall correlation of solar radiation intensity and surface temperature was good. Therefore, the current model is considered to be within the acceptable range for subsequent analysis and evaluation.

3.4. Simulation results

The solar radiation intensity and thermal stress were obtained at different heights using the simulation model to explore their variation patterns with height on the statue surface.

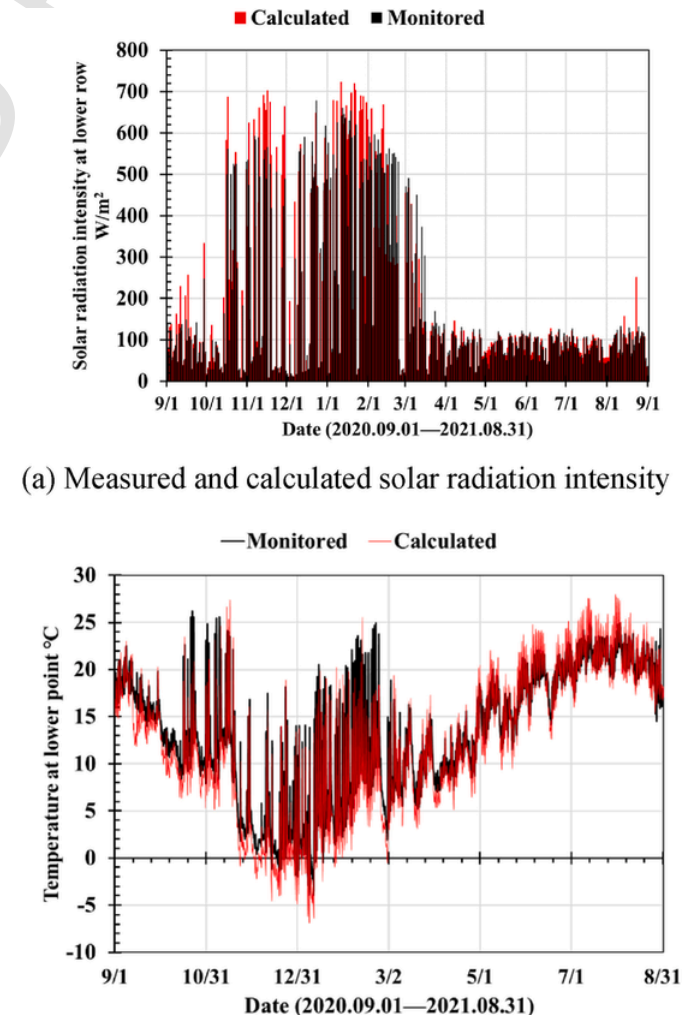


Fig. 7. The model validation (the calculated and measured results at the lower point).

3.4.1. Solar radiation

Because of the orientation (15° south by east) and the ruined eaves (uneven shades) in Grotto No. 3, the change in solar radiation is complex. Therefore, simulation is necessary to determine the spatiotemporal distributions of solar radiation on the statue surface.

The calculated results included 19 heights on the statue surface, with H10, H80, H130, H190 and H200 being used for further comparison. It can be seen from Fig. 8 that higher points tend to receive lower solar radiation intensity with a shorter influenced period (delayed insolation), which is in agreement with the changes observed in the results of monitoring (Section 3.2, B). H10 represented the points under long-term exposure to direct insolation; H200 represented those often in the shadow of the eaves; H190 represented the points which shift repetitively between the shadow and the sunshine area; H80 and H130 (at middle heights) were chosen because they reveal the changes at the beginning or the end (Fig. 9a and c) of the influenced period (October or March), despite being mostly identical to H10 during the influenced period (Fig. 9b).

Table 3 shows that the maximum intensity, the amount of solar radiation, and the hours of direct sunlight all decrease as the points become higher. There were 469.7 h of direct sunlight with solar radiation intensity over 150 W/m² at H10, 436.3 h at H80, 371.3 h at H130, 161.5 h at H190, and only 29.7 h at H200. As to the amount of radiation, a total of 298.7 kW h/m² was obtained at H10, and 290.3, 275.2, 209.6, and 181.1 kW h/m² at H80, H130, H190 and H200, respectively. Significant decreases in both hours of direct sunlight and total amount were

found with increasing height. It can also be seen that the solar radiation intensity of the lower points (Fig. 9) retained large values for longer durations. Moreover, the radiation intensity of higher points tends to exhibit a temporary decrease and rebound because of shadow from the grotto eaves, forming a bimodal curve (H190 and H200 on 23rd December).

3.4.2. Thermal stress at different heights

Using the simulated surface temperatures and the parameters of the clay (Table 2) in Equation (9), the thermal stress of the statue surface (YOZ plane, Fig. 4) can be acquired.

Fig. 10a shows the annual changes in thermal stress at the five points. The compressive stress is mostly found from October to May, and the tensile stress tends to occur during the rest of the year. However, under the influence of solar radiation, the frequency of tensile stress increases, being superimposed with compressive stress during the influenced period. Throughout the year, the proportions of compressive and tensile stress differ, with a higher frequency of tensile stress (57.7%; an average of the five heights). This may be related to heated surface temperatures under direct insolation during the influenced period. Additionally, the changes in thermal stress are similar to the fluctuation curves of surface temperature in Fig. 7b, showing a spatiotemporal distribution at different heights.

Specifically, larger variations and maximums were found during the influenced period and at the lower points. The largest daily ranges at H10, H80, H130, H190 and H200 were 28.2, 28.1, 28.2, 24.6, 12.5 kPa, respectively. As listed in Table 4, the maximum values of compressive and tensile stress also showed similar changes with height. The maximum tensile stress values at H10, H80 and H130 were much larger than those at H190 and H200, meaning lower points experienced more extreme exposure under the ruined eaves. The differences among H10, H190 and H200 reached nearly 40 kPa in the middle of the influenced period (Fig. 10a), while there is almost no difference (i.e., a difference of less than ± 5 kPa) among the five heights for the rest of the year. The maximum compressive stresses also increased with decreasing height, but the values were smaller than those in tensile stress, and only tiny differences in height were shown. Taking the winter solstice as an example (Fig. 10b), a repeated diurnal pattern in stress variation was found, where thermal stress at lower points tended to increase earlier than that at higher positions (21st December), and remain at greater values during the day. Additionally, multiple peaks of tensile stress could occur during a single day (23rd December).

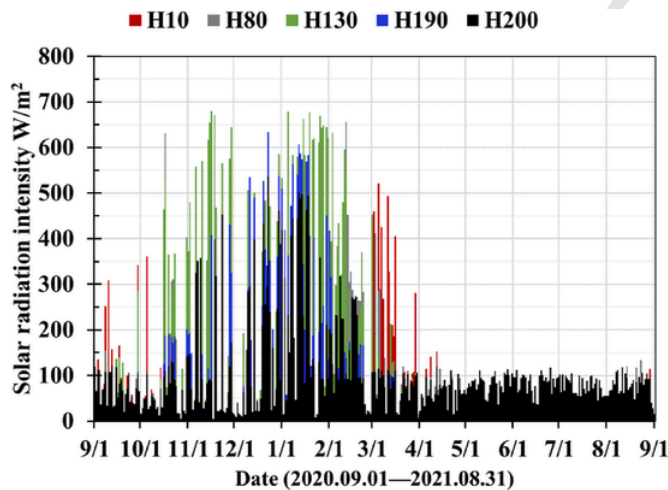


Fig. 8. Calculated solar radiation intensity at different heights.

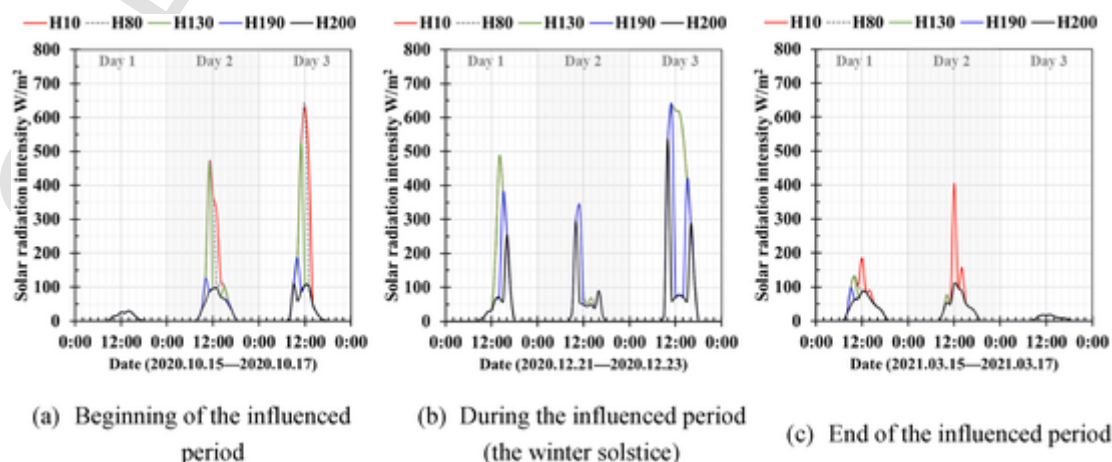


Fig. 9. Solar radiation at different heights and periods.

Table 3
Solar radiation at different heights.

Position		H10	H80	H130	H190	H200
Range of Solar Radiation Intensity	W/m ²	0–680.4	0–680.4	0–680.4	0–633.4	0–535.9
Amount of Solar Radiation	kW·h/m ²	298.7	290.3	275.2	209.6	181.1
Hours of Direct Sunlight	h	469.7	436.3	371.3	161.5	29.7

4. Discussion

4.1. Solar radiation and thermal stress

The annual variation in the solar altitude is the key factor affecting the amount and intensity of solar radiation, and sometimes that of surface weathering [62]. When the solar altitude falls below a certain angle, the statues in the two rows successively lose the shading offered by the eaves, resulting in different durations and amounts of solar radiation. However, the solar radiation intensities at different heights reach their peaks on the same day, meaning that the shading effect of the ruined eaves is insufficient for both rows. Meanwhile, during the rest of the year, higher solar altitudes guarantee a larger shading area; thus, mainly diffuse radiation occurs. Taking the maximum hourly temperature changes over the course of a year as an example, it is only 10.75 °C at H200, close to 9.08 °C in the ambient air, while lower points (H190 to H10) range from 21.39 to 21.49 °C. In comparison to the ambient air temperature, the significant influence of solar radiation on increases in surface temperature can be confirmed. Throughout the year, the larger the temperature changes under direct insolation, the larger the thermal stress on the statue surface. The relationship between the solar radiation intensity (Fig. 8) and the thermal stress (Fig. 10a) was established.

Moreover, the correlation between them is also characterized by a diurnal pattern (as mentioned in the literature [10]). For example, during the daytime, tensile stress could appear or increasing rapidly after direct insolation, in different manners from top to bottom. When direct sunlight at some points (H190) was lost and then regained (Fig. 9b), the thermal stress showed a corresponding decreasing and then increasing trend (Fig. 10b). The stress differences between different heights generated in the daytime gradually decreased and tended to be uniform until the next insolation exposure.

4.2. Thermal stress and suggested threshold

Many standards have set limits to the fluctuations in temperature and relative humidity for the conservation of cultural properties [63–66]. Proposing a threshold or its determination method provides scientific guidance for the protection of heritage sites. The failure of one material is generally caused by strength or deformation exceeding that

of its allowable value. In engineering practices, strength theory is often used to determine the failure of materials, that is, the maximum tensile stress, maximum tensile strain, maximum shear stress and strain-energy density. Therefore, the threshold of thermal stress will be determined on the basis of the mechanical strength of the material.

In this case, considering the damage of flaking on the statue surface and much greater compressive strength of clay than its tensile one (tensile failure occurs first), the theory of maximum tensile stress was selected as a criterion of its failure caused by thermal stress, that is, the tensile strength of clay from 10 to 80 kPa [67–69]. To ensure the safety of cultural heritages, the minimum tensile strength was selected as a reference threshold for assessing the damage risk caused by thermal stress, namely 10 kPa for clay. However, the exceedance of higher strength values was also considered.

Table 5 lists the situations in which the thermal stress exceeded the suggested threshold and larger values in a year. The frequencies with which the suggested threshold (10 kPa) was exceeded were 3122, 3127, 3077, 2867, and 2819 times at the positions from H10 to H200, respectively. The frequency with which stress exceeded 10–45 kPa was different at each point, but lower points usually suffered from solar-induced thermal stress the most. The frequencies at H80 were slightly higher than H10, which may be related to the location being in the corner and the different materials near H10 (bottom of the lower row in Fig. 3c), disadvantaging heat transfer among adjacent points. Moreover, it was confirmed by a statistic that the frequency difference in heights is mostly found during the influenced period. Taking H10 and H200 as an example, the frequency at H10 has 348 more counts than that of H200 from October to March, while 45 counts less for the rest of the year. Similar results are also found at other heights especially in the upper row of statues (over H110), that is, the frequency difference between one point and H10 increases with height.

To explore the short-term influence of solar radiation, efforts have also been made by applying temperature changes from T_0 to T_1 over different time spans (ranging from 5 min to 24 h) to the ΔT . Under these circumstances, it can be observed that the threshold could be exceeded within 5 min (22 times at H10), and the frequency increases and then decreases as the time span becomes larger. The results indicate that short-term temperature changes under direct insolation are big enough to threaten the conservation of painted statues.

Considering the loss rate distribution of both the paint and earthen plaster layers (Fig. 5) and the calculation results (Sections 3.4), it can be deduced that the deterioration is more likely to occur on surfaces that are frequently exposed to sunlight. The investigation into the deterioration of the statues and the thermal stress calculated in this study support the correlation evaluation of flaking on painted statues under direct insolation and help to establish an early warning reference for thermal stress and control of radiation intensity.

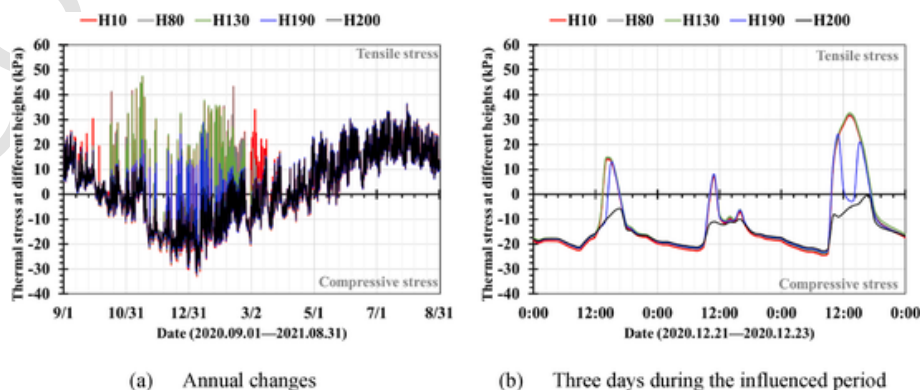


Fig. 10. Thermal stress at different heights.

Table 5

Frequencies with which thermal stress exceeds the different levels of strength.

Value of σ_T	H10	H80	H130	H190	H200
>10 kPa	3122	3127	3077	2867	2819
>15 kPa	2021	2015	1935	1753	1783
>20 kPa	988	981	940	830	819
>25 kPa	371	380	362	299	277
>30 kPa	82	84	78	39	30
>35 kPa	21	23	21	1	1
>40 kPa	13	13	8	0	0
>45 kPa	1	1	1	0	0

Table 4

Compressive and tensile thermal stress.

Position	Compressive stress ($\sigma_{\min}$; absolute value)	Tensile stress ($\sigma_{\max}$)
	kPa	
H10	32.9	46.6
H80	32.5	47.0
H130	32.2	47.1
H190	32.0	36.4
H200	31.2	35.8

4.3. Preservation conditions under different protection measures

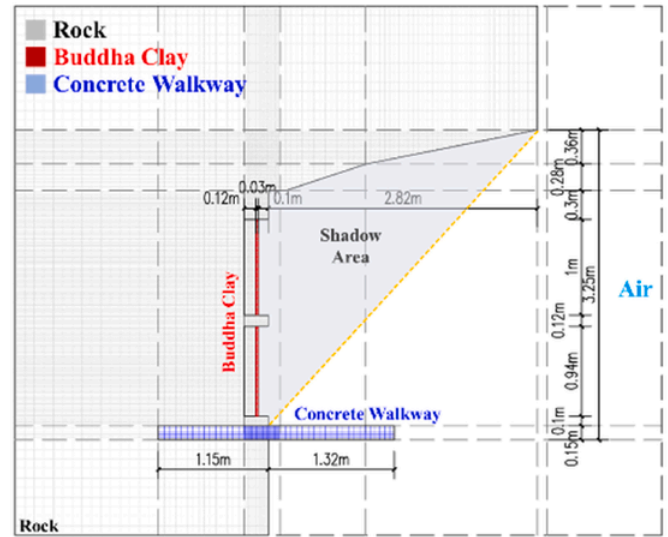
Based on the correlation analysis between thermal stress and the deterioration of the statue surfaces in Grotto No. 3, Maijishan Grottoes, two hypothesized protection measures were introduced and compared with the current situation (Case I, Fig. 3b). To eliminate the impact of direct insolation, Case II assumes a hypothetical shading facility, as a result of which the statue is in shadow after the sunrise (Fig. 11a). Case III (Fig. 11b) is a restoration of Grotto No. 3 based on architectural historical research performed by Fu Xinian [70].

The model parameters in Case II were set to be consistent with those in Case I, but with the assumption of there being zero direct solar radiation on the statues. The influence of introducing the facility on air change, wind speed, etc., are not considered, i.e., no specific form is considered. In Case III, the section of its inferred original form, where the window openings are found, was considered in the simulation model. The parameters used in Case III were modified, where the air inside and outside the grotto is distinguished, with an air change rate of 10 times per hour. The comparison of the three cases covers both solar radiation and thermal stress, and lower positions (H10, H80 and H130) will be our focus, as the numbers of event exceeding the threshold did not increase significantly at points higher than H130 under the protections in Case II and III.

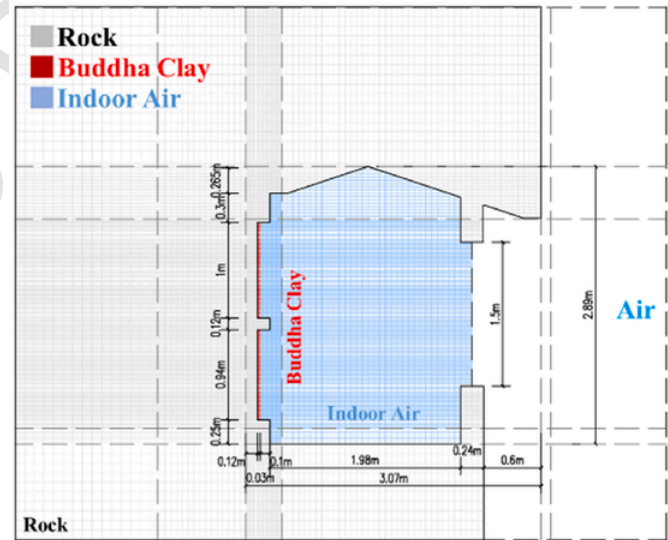
In terms of solar radiation, the spatiotemporal distribution was also found in Cases II and III, with greater values tending to occur at lower points. Specifically, the durations of direct sunlight in Case I and Case III were 469.7 and 249.8 h at H10, 436.3 and 81.2 h at H80, 371.3 and 10.2 h at H130, with a decrease in 46.8%, 81.4%, and 97.3%, respectively, while there is almost no sunlight over H160 in Case III. In Case II, the statues were in shade all year round. The total amount of radiation in Case III, compared with Case I, are 147.9, 89.1, and 63.8 kW h/m² at H10, H80, and H130, with a decrease in 50.5%, 69.3%, and 76.8%, respectively, and those higher than H160 are almost same (58.5–59.0 kW h/m²). The decrease in Case III is obvious, but a smaller shading effect was provided at lower points (both in the duration and the amount of solar radiation).

In Case II, all points received about only 173.1 kW h/m² (diffuse radiation). Therefore, both Case II (shading facility) and Case III (restored state) effectively reduced the time and the amount of solar radiation.

Thermal stress at H10, H80 and H130 in the three cases could be observed (Fig. 12a–c). Throughout the year, the thermal stress in Case I was the greatest (maximum over 46.6 kPa) and that in Case III was the



(a) Case II: Hypothetical shading facility

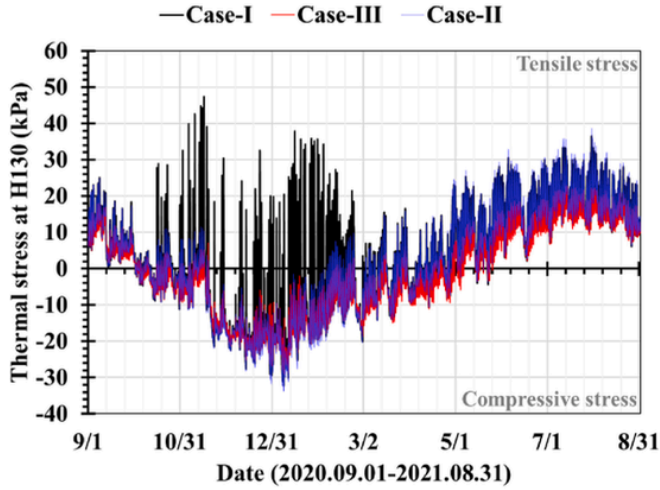


(b) Case III: Restoration

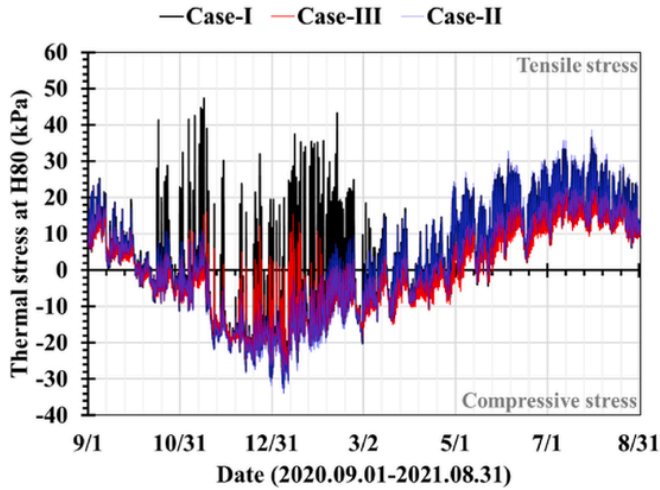
Fig. 11. Different protection measures.

lowest (23.4–36.4 kPa) among the cases at all three heights. Stress exhibited a rapid decline with increasing height in Case III, which is in agreement with the distribution of solar radiation described above. The maximum values of the tensile stress from H200 to H70 were similar (22.8–23.7 kPa). As height decreased, the maximum became larger. It varied from 25.8 to 27.5 kPa between H60 and H30 and exceeded 30.0 kPa below H20. It is assumed that the restored structure in Case III provides a transition space, buffering the fluctuations of the outside environment, thus resulting in less thermal stress. Both the extremes and fluctuations of the inside temperature were smaller than those on the outside, that is, the minimum and the maximum were about 4.1°C higher and 11.1°C lower than those of the outside, and the variation range was only about 71.2% of that of the outside.

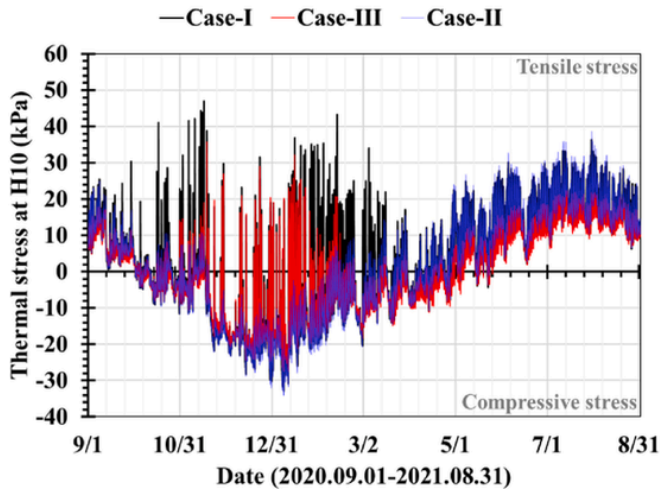
For events exceeding the threshold, the frequencies in different cases are shown in Fig. 13. It is clear that the frequencies decrease successively from Case I to III. As there was no sunlight in Case II, the ex-



(a) H130



(b) H80



(c) H10

Fig. 12. Thermal stress at different heights in the three cases.

ceedances at all heights were similar and fluctuated within a hundred times. Compared with the calculation conditions in Case I, it is speculated that this level of exceedance is mainly brought about by the influence of seasonal change of outdoor temperature, rather than direct solar radiation. Therefore, the difference in frequency between Case II and the other two cases reflects improvement or negative effects under corresponding conservation conditions. In Case I, the frequency exhibited a clear decrease with increasing height, which was also superimposed with a change that the points away from the corner (for example H10 and H90, Fig. 3c) will show exceedance more frequently due to better heat transfer with adjacent points. The increments compared with Case II implicate the influence of direct insolation and insufficient protection under the ruined eaves. The situation was greatly reduced in Case III, with a decrease from 18.7% to 26.3% at different heights. The difference among the heights was observable, decreasing from 354 times in Case I to 137 times (over 61.3%).

These results show that both Case II and III are able to reduce the risk of thermal stress. The protection effect in Case III (restoration based on historical research) is the most obvious due to the buffering effects of the transitional space, while the protection afforded in Case II (hypothetical shading facility) is limited. Either a buffering space or a shading facility that can cover the entire surface of the Buddha is effective and necessary. However, the design of the facilities should incorporate several aspects, including protectivity, the feasibility of installation, and the impact after installation (i.e., appearance, etc.).

4.4. Application of the threshold

The proposed threshold of thermal stress may be of guiding significance for managers of heritages when performing monitoring and early warning. Combining the simulations of the three cases in Section 4.3, it can also provide a reference for the development and implementation of protective measures.

As shown in Fig. 14, a monitoring and early warning system for thermal stress can be integrated into an existing monitoring system or established, which includes three modules. The real-time hygrothermal monitoring of solar radiation intensity, ambient air temperature and relative humidity provides boundary conditions for the simplified one-

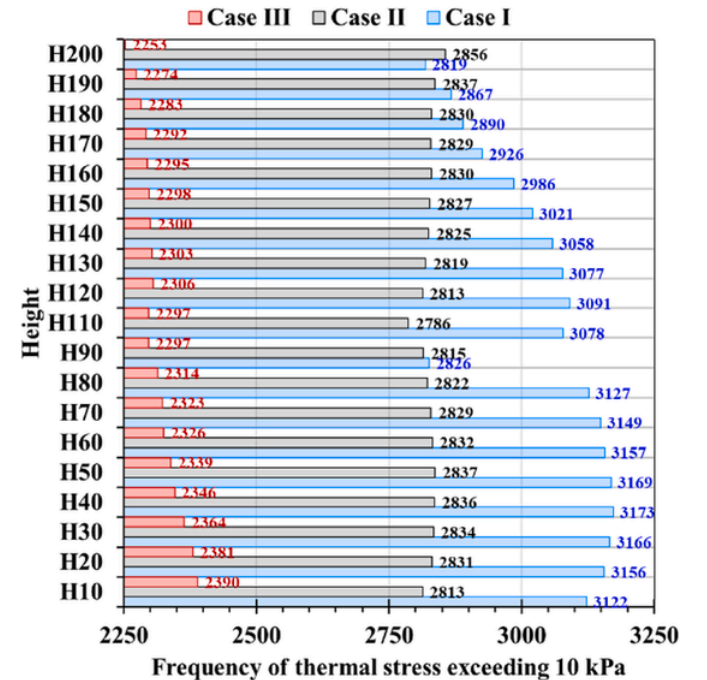


Fig. 13. Exceeding threshold events at different heights in different cases.

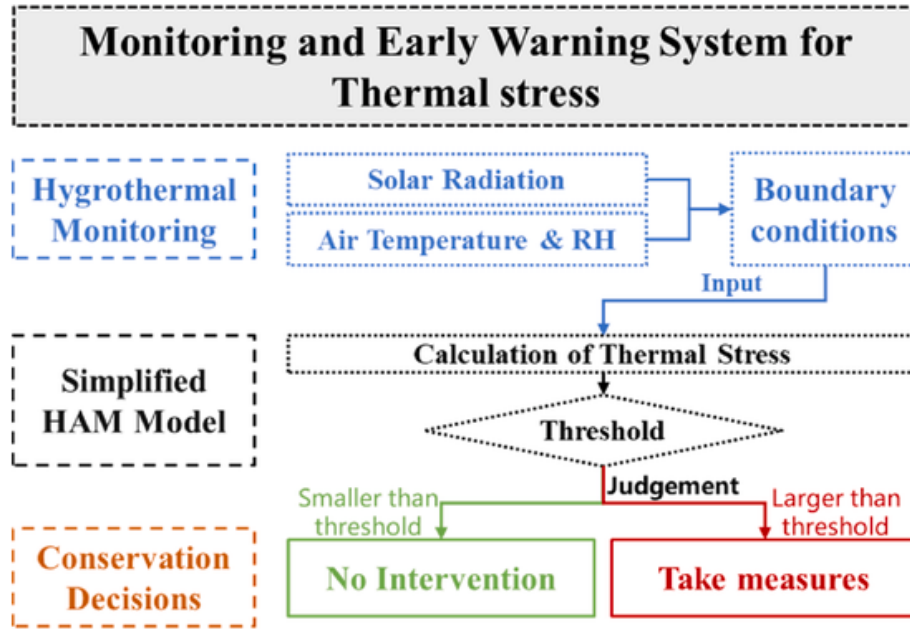


Fig. 14. Monitoring and early warning system.

dimensional model (on the basis of Section 2.3). Then, the model outputs calculated thermal stress and it is compared with the proposed threshold. Possible protection measures should be taken when the stress exceeds the threshold, while no intervention otherwise.

In addition, it can be achieved by a simpler method when budget is limited. For example, by transforming the equation for the thermal stress (Equation (9)) into one for the temperature change (Equation (10)), the monitoring index becomes easier to be achieved, that is, the changes in surface temperature. On the basis of the suggested threshold of tensile stress (10 kPa) and the reference values of the material presented in Table 2, the threshold of the surface temperature increase can be deduced, which was 6.0 °C.

$$\Delta T = \frac{\sigma_T \cdot (1 - \nu)}{E \cdot \alpha} \quad (10)$$

It is known from the simulation of the two hypothetical situations in Section 4.3 that the thermal stress is inevitable under the temperature fluctuation throughout a year, meaning that the aging of heritage materials in the outdoor environment is continuous with a certain rate. Therefore, efforts should be made to decrease the influence of external factors, reducing the frequencies of materials exceeding their strength as much as possible. For the heritage managers, their main objective should be to reduce the temperature variation of the statue surface (creating more stable preservation environment). For example, the shading facility (Case II) could reduce the impact of long-term or periodic solar radiation. And creating a transition space (Case III) buffers the influence of both seasonal temperature changes and that of the solar radiation. However, the specific implementation of protective measures should be fully justified and comply with local policies and laws.

5. Conclusion and future works

By means of a field survey, in this paper, Grotto No. 3 was monitored, and a heat and moisture transfer model was established to simulate and analyze the relationship between its deterioration and the environment, especially the influence of direct insolation. As a result, this research provides data for the correlations between surface deterioration and changes in temperature and radiation. On this basis, an early warning threshold for thermal stress was proposed as a reference in heritage conservation. Moreover, three preservation models for the grotto

(current situation, current situation with hypothetical shading, and restoration) were compared to evaluate the protection afforded by possible shading facilities. The conclusions can be summarized as follows.

- (1) Direct insolation has a significant effect on changes in the surface temperature of the statues, thus leading to high thermal stress. Solar radiation with longer duration and greater intensity showed a positive correlation to greater degrees of surface loss. This is a common but urgent problem faced by all cultural heritages in open or semi-open environments.
- (2) Evaluation methods for thermal stress provide an effective means of performing data analysis and achieving early warnings in heritage conservation. It is feasible and propagable in practice to establish an early warning system on the basis of the threshold of temperature change for artifacts at cultural heritage sites. This threshold was suggested to be 6.0 °C, according to the tensile strength (10 kPa) and the characteristics of the clay.
- (3) The length of the cornice plays an important role in protecting the statues in the grottoes. The thermal stress on the surface of the statue under the eaves increased from top to bottom (height), and the frequency with which the threshold was exceeded increased accordingly. The managers of heritage sites should scientifically evaluate the protective effect on each statue in order to delay deterioration.
- (4) For heritages in open or semi-open environments, it is necessary and effective to adopt protective facilities for protection, but the impact of such facilities on the statues inside, through ventilation, thermal capacity, etc., needs to be further examined and evaluated.

Though, it should be acknowledged that there are three aspects of limitation in current research, thus improvements will be made regarding these issues in the future. First, the spatial distribution of thermal stress in depth and the distribution caused by color difference (temperature becomes higher on darker pigments) were not involved. Second, the moisture content stays at low level (0.042–0.067 m³/m³; saturation: 8.1–12.9%), thus the influence of moisture expansion and salt crystallization might be small and were not considered here. Third, due to limitation of testing conditions (heritage sites), the parameters of materials are mainly obtained from references. Therefore, further property tests,

experiments and improvement of the model will be conducted in the subsequent study to obtain the stress distribution and realize more accurate risk assessment.

CRedit authorship contribution statement

Ruobin Wu : Writing – original draft, Software, Methodology, Formal analysis, Data curation. **Yan Ma** : Visualization, Formal analysis. **Huarong Xie** : Validation. **Xudong Wang** : Funding acquisition, Supervision, Writing – review & editing. **Shuichi Hokoi** : Writing – review & editing, Supervision, Methodology. **Yongqiang Yue** : Resources, Project administration. **Junjian Hu** : Resources. **Baogang Mu** : Methodology, Investigation. **Yonghui Li** : Writing – review & editing, Supervision, Project administration, Investigation, Funding acquisition, Data curation, Conceptualization.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data availability

The data that has been used is confidential.

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Appendix A. Nomenclature

μ	water chemical potential, $[J/kg]$
R_v	gas constant of water vapor, $[Pa \cdot m^3 / (kg \cdot K)]$
T	temperature, $[K]$
RH	relative humidity, $[-]$
c_p	specific heat capacity, $[J / (m^3 \cdot K)]$
t	time, $[s]$
λ	thermal conductivity, $[W / (m \cdot K)]$
r	heat of phase change in water, $[J/kg]$
λ'_{Tg}	water vapor conductivity under the gradient of temperature $[kg / (m \cdot s \cdot K)]$
$\lambda'_{\mu g}$	water vapor conductivity under the gradient of water chemical potential $[kg / (m \cdot s \cdot (J/kg))]$
ρ_{water}	density of water, $[kg/m^3]$
Ψ	volumetric moisture content, $[m^3/m^3]$
λ'_T, λ'_μ	volumetric moisture content, $[m^3/m^3]$
λ'_T, λ'_μ	water (liquid and vapor) conductivity under the gradient of temperature $[kg / (m \cdot s \cdot K)]$ and water chemical potential $[kg / (m \cdot s \cdot (J/kg))]$
$\vec{n}$	unit vector in the direction of gravity, $[-]$
g	gravitational acceleration, $[= 9.8 m/s^2]$
J	direct solar radiation (of the earth surface, $[W/m^2]$
J_0	solar constant, $[= 1367 W/m^2]$
P	equivalent factor considering the effects of weather conditions (clouds, atmospheric transmission, etc.), $[-]$
θ	solar altitude, $[rad]$
J_{sh}	diffuse radiation, $[W/m^2]$
h_c	convective heat transfer coefficient, $[W / (m^2 \cdot K)]$
a	constant (depending on the state of the plane). Vertical: $a = 2$; horizontal: $a = 2.5$ (when heat flows from down to up), and $a = 1.3$ inversely
$T_{surface}$	surface temperature of solid materials, $[K]$
T_{out}	outdoor air temperature, $[K]$
v_{wind}	velocity of the wind, $[m/s]$
h_r	radiative heat transfer coefficient, $[W / (m^2 \cdot K)]$
$\overline{\Psi}_{12}$	average radiation angle factor of object 1 versus object 2, and vice versa ($\overline{\Psi}_{21}$); In this model: $\overline{\Psi}_{12} = \overline{\Psi}_{21} = 1 C_{1,2}$
	average radiation angle factor of object 1 versus object 2, and vice versa ($\overline{\Psi}_{21}$); In this model: $\overline{\Psi}_{12} = \overline{\Psi}_{21} = 1 C_{1,2}$
	equivalent radiation coefficient between object 1 and object 2, $C_{1,2} = \frac{C_1 \times C_2}{C_b}$, $C_b = 5.68 W / (m^2 \cdot K^4)$ is the radiation coefficient of the absolute black body, $C_1 = 5$ and $C_2 = 5.4$ (empirical values)
h_{total}	comprehensive heat transfer coefficient, $[W / (m^2 \cdot K)]$
α	linear thermal expansion coefficient, $[1/^\circ C]$, or $[(m/m) \cdot K^{-1}]$
ϵ	thermal strain, $[-]$
E	Elastic Modulus, $[Pa]$
ν	Poisson's ratio, $[-]$
σ_T	thermal stress, $[kPa]$

Appendix B. Hygrothermal Properties

The hygrothermal properties of the materials used in the simulation are based on the literature [51]. The specific heat capacity and porosity are listed in Table A. The thermal conductivity, volumetric moisture content, and moisture conductivity under the gradient of water chemical potential are shown in Figure A, Figure B, and Figure C, respectively. The moisture conductivity under the gradient of temperature is shown in Figure D.

Table A
Specific heat capacity and porosity

Material	Specific heat capacity (c_p), [J/(m ³ ·K)]	Porosity, [–]
Rock	2590 × 840	0.26
Clay	1500 × 850	0.52

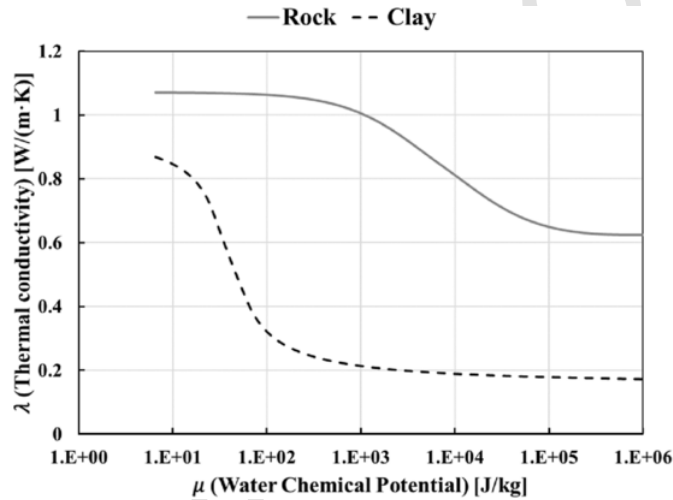


Fig. A. Thermal conductivity and water chemical potential.

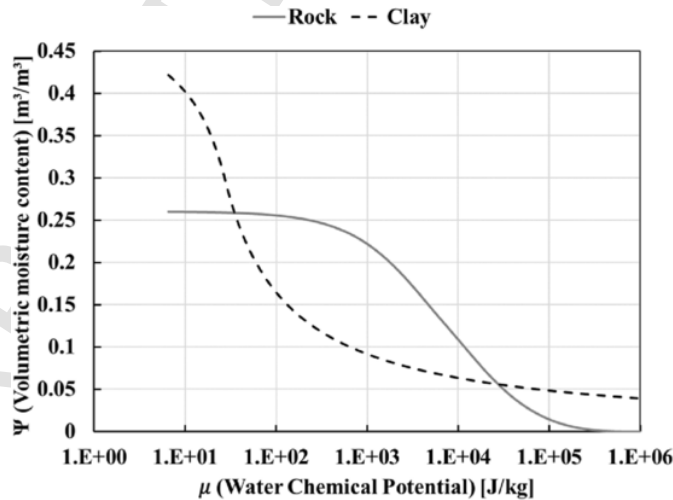


Fig. B. Volumetric moisture content and water chemical potential.

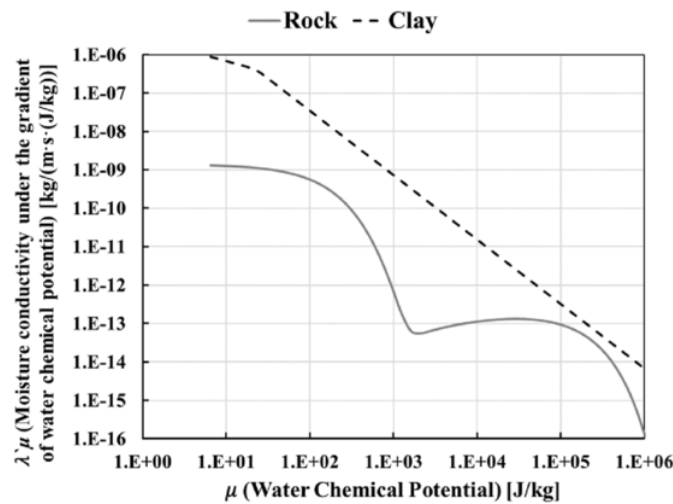


Fig. C. Moisture conductivity under the gradient of water chemical potential.

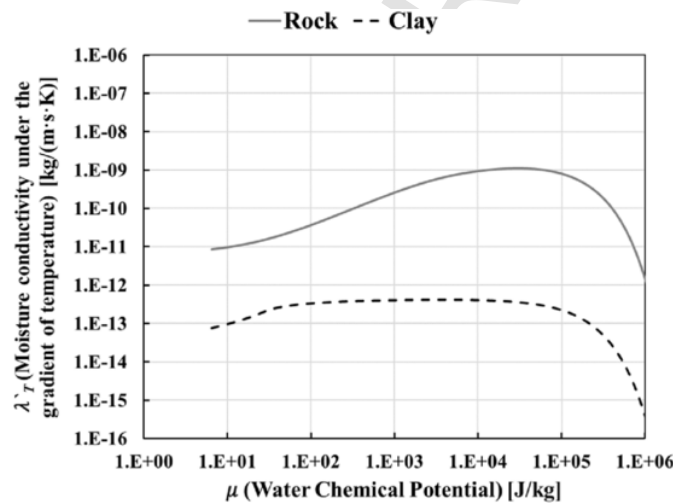


Fig. D. Moisture conductivity under the gradient of temperature.

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